Object Identification and Recognition (IV)

Introduction to Computational and Biological Vision

CS 202-1-5261

Computer Science Department, BGU

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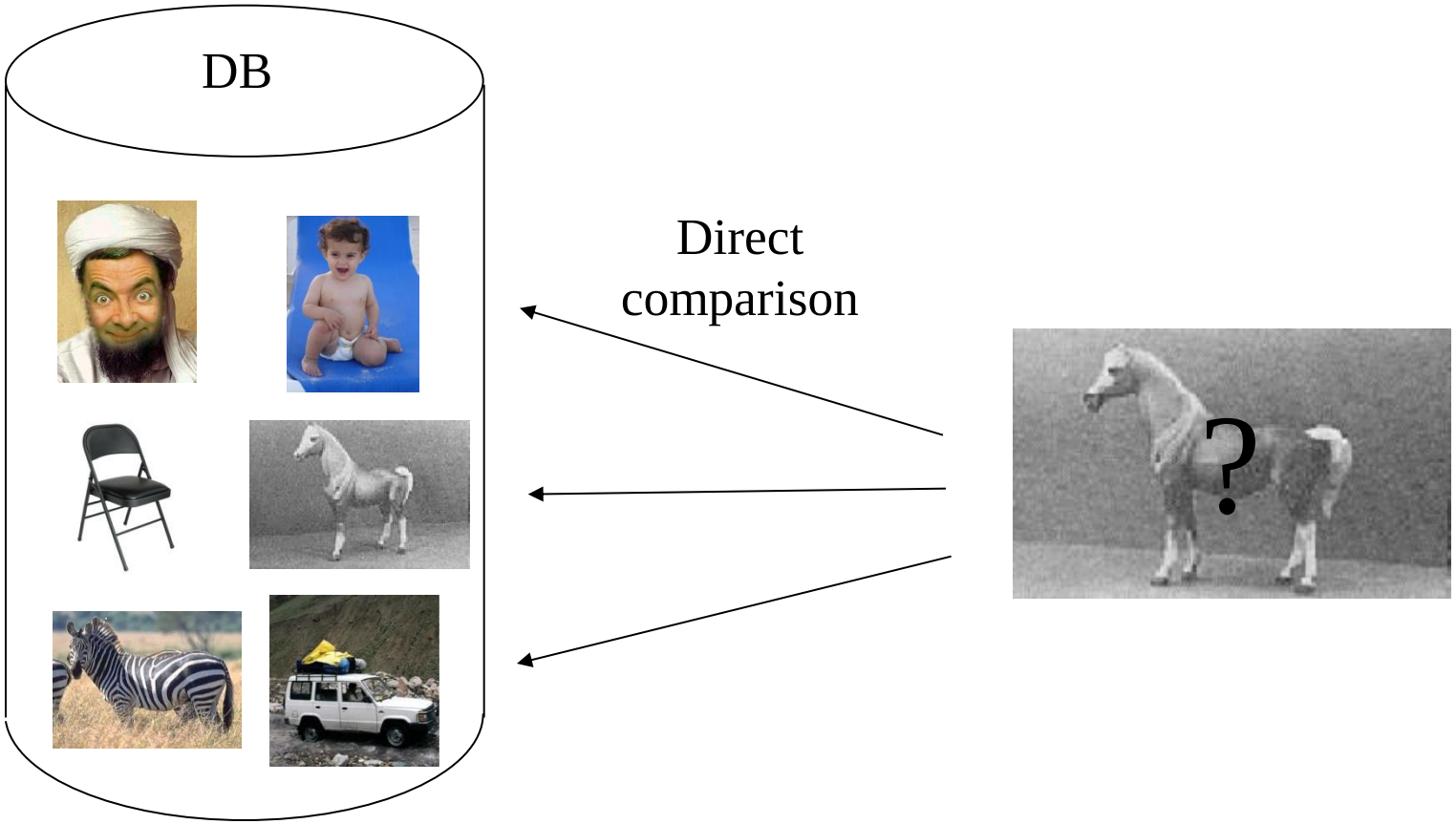
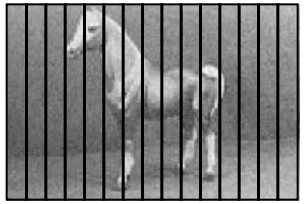
Appearance-Based Recognition

Recognition by appearance matching



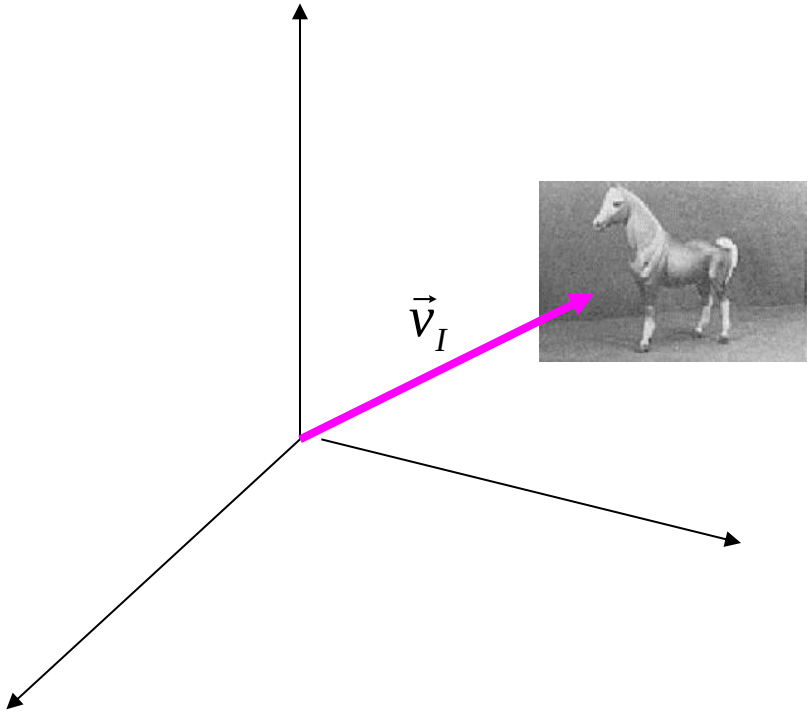
Appearance-Based Recognition

Abstraction - images as vectors



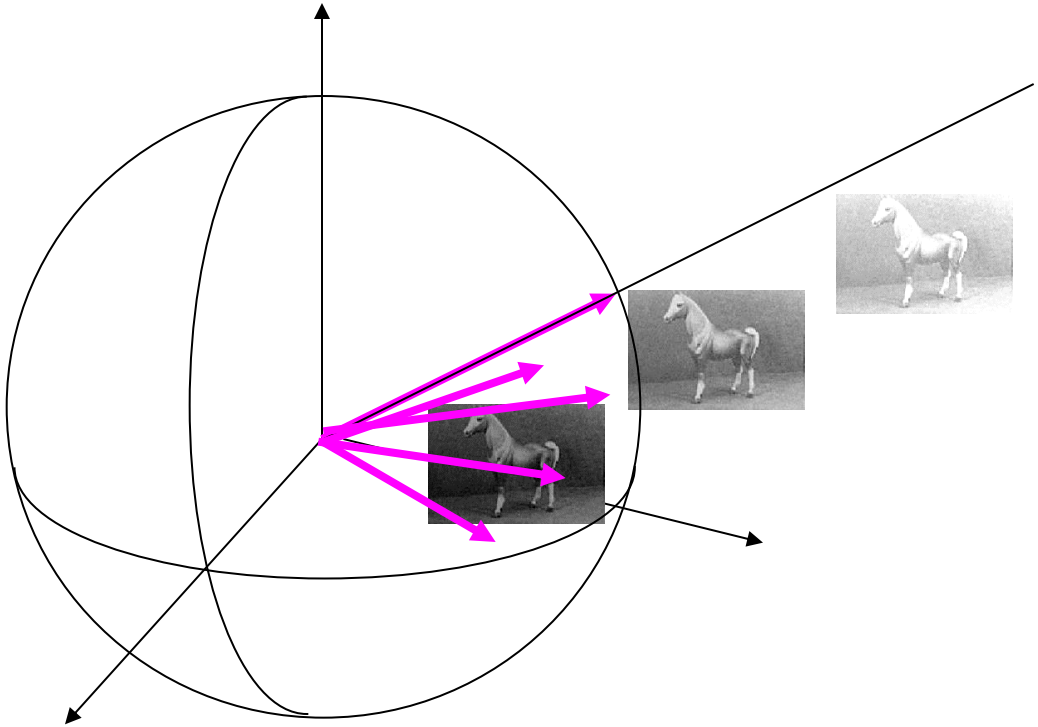
$\Rightarrow I$

$\Rightarrow \vec{v}_I$



Appearance-Based Recognition

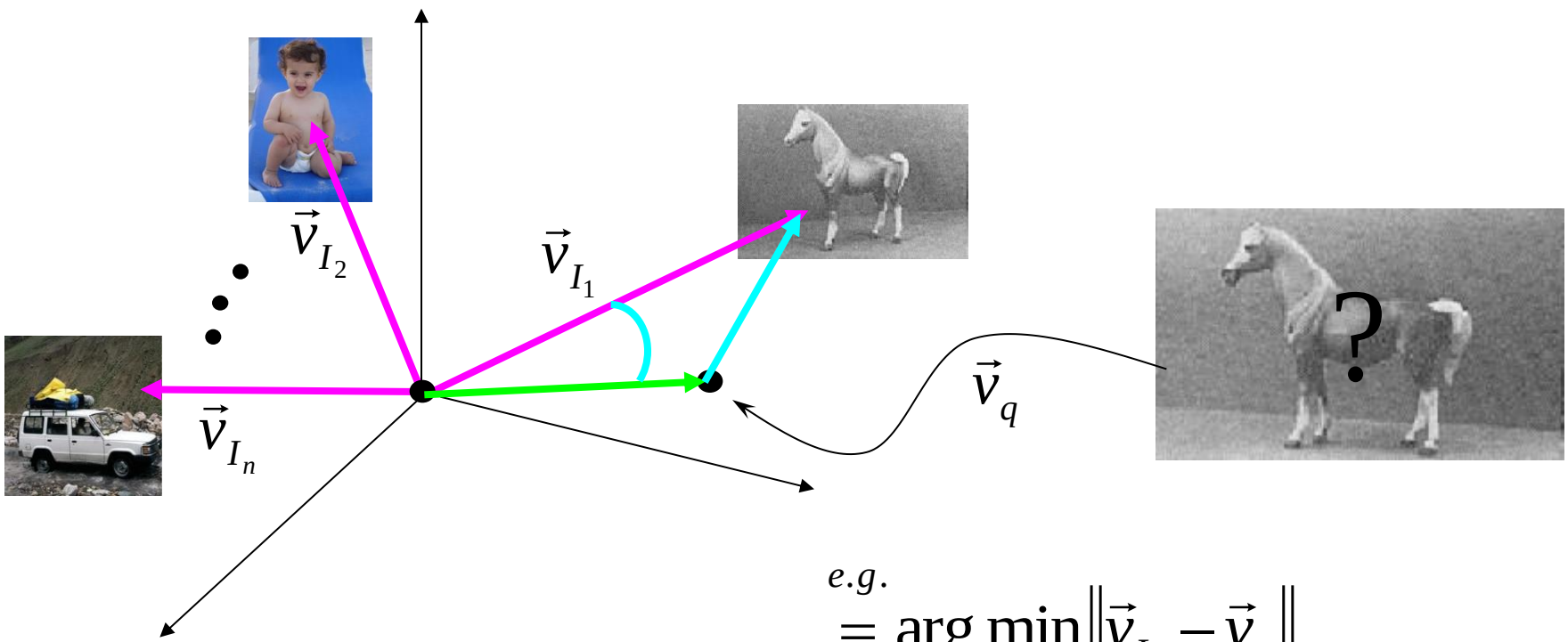
Abstraction - images as vectors



Normalized images $\left\|\vec{v}_I\right\| = 1$

Appearance-Based Recognition

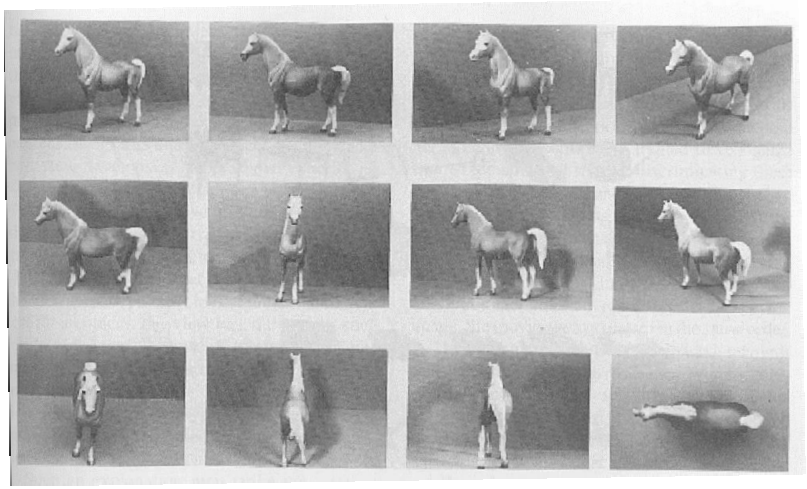
Recognition by appearance matching

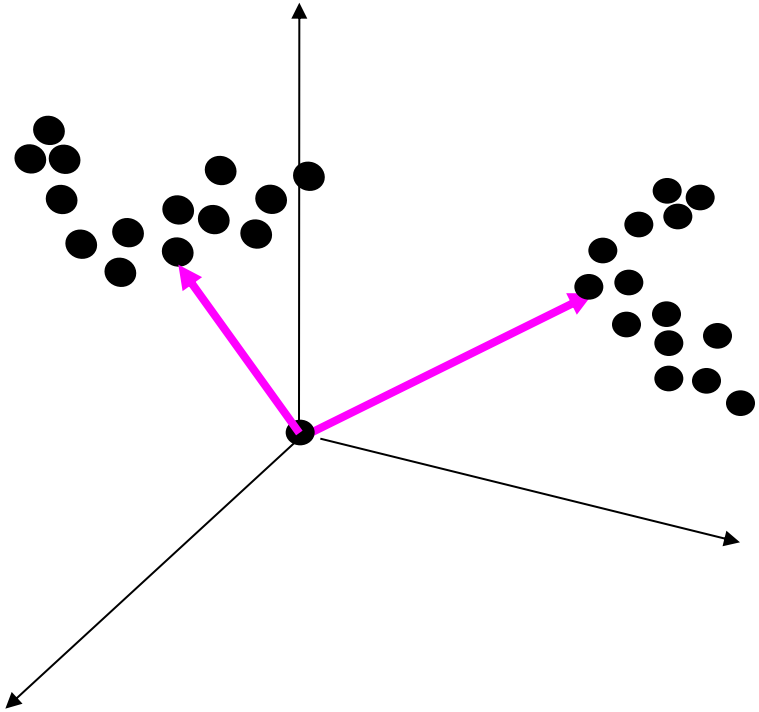


$$\arg\min_i D(\vec{v}_{I_i}, \vec{v}_q) \overset{e.g.}{=} \arg\min_i \left\| \vec{v}_{I_i} - \vec{v}_q \right\| \overset{e.g.}{=} \arg\min_i \left(\hat{v}_{I_i} \cdot \hat{v}_q \right)$$

Appearance-Based Recognition

Problems in naive appearance matching

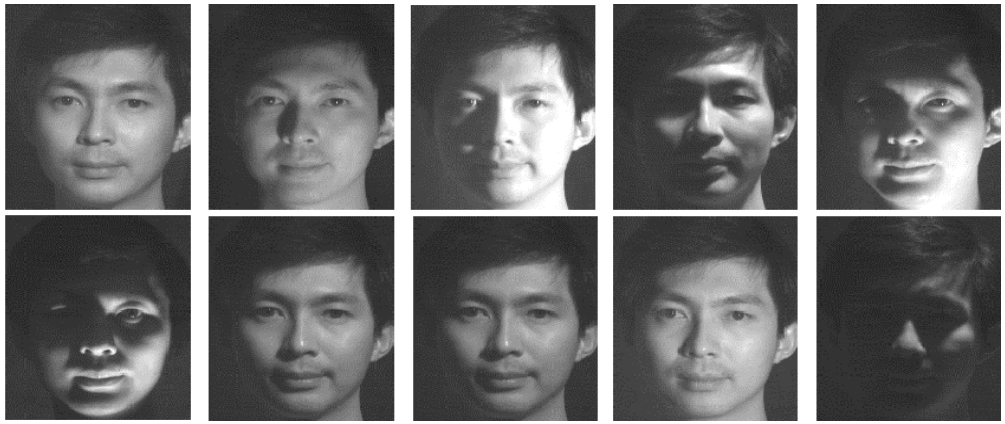


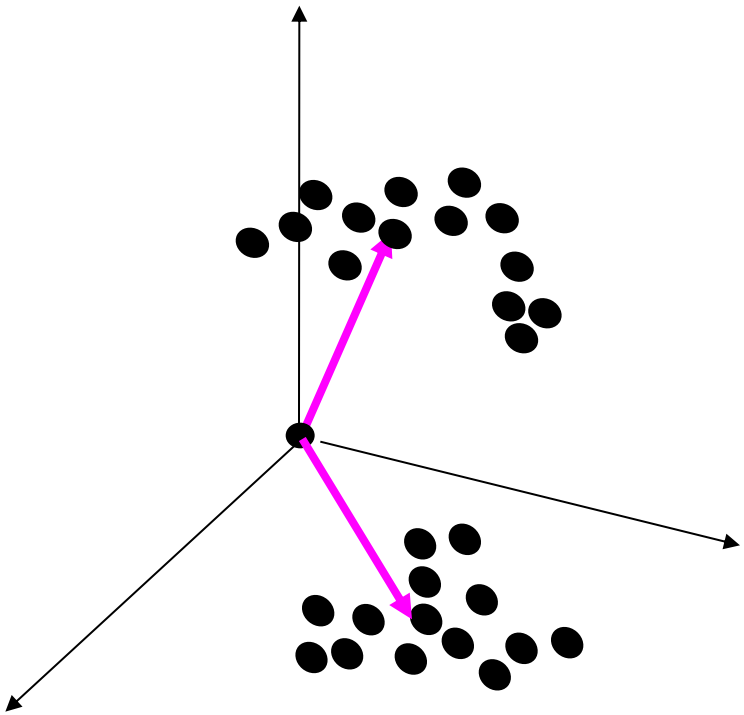


Variability due to pose/viewing

Appearance-Based Recognition

Problems in naive appearance matching





Variability due to illumination

Appearance-Based Recognition

Problems in naive appearance matching

Image size N x N pixels, B bytes/pixel

N_V viewing directions per object

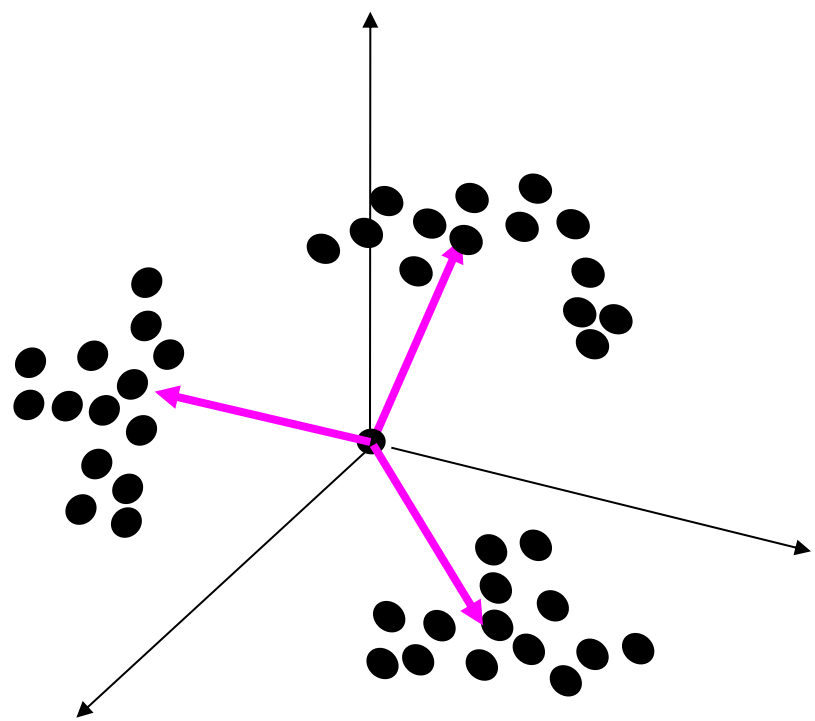
N_L illumination directions per object

N_o objects in database

$$\Downarrow$$

$$N_o \cdot B \cdot N^2 \cdot N_V \cdot N_L \quad bytes$$

$$\boxed{\begin{array}{l} N = 256 \\ B = 3 \\ N_V = 64 \\ N_L = 64 \\ N_o = 100 \end{array} \quad \Rightarrow \quad 75 GBytes}$$



Cost of representation

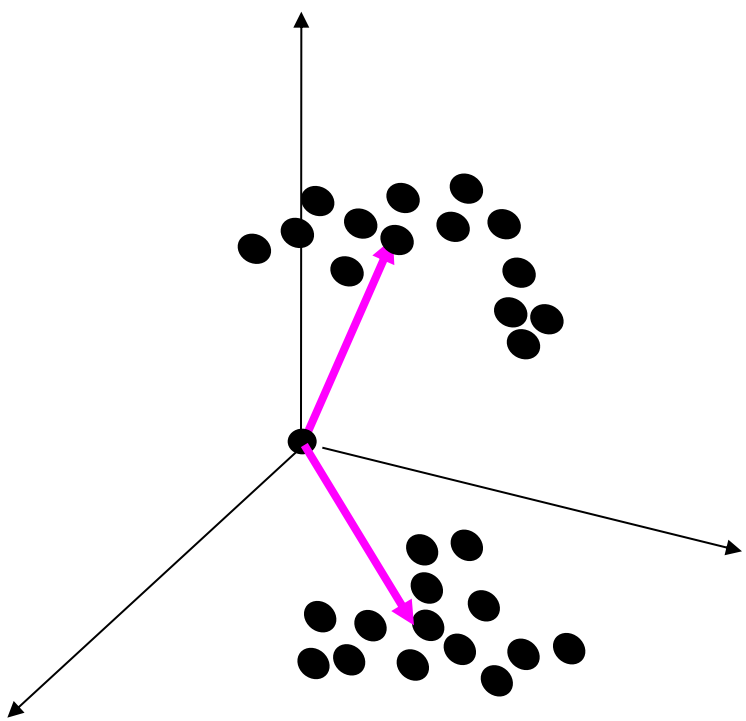
Appearance-Based Recognition

Problems in naive appearance matching

$$\left\| \vec{v}_{I_i} - \vec{v}_q \right\| = \sqrt{\sum_{k=1}^{N^2} \left(\vec{v}_{I_i}(k) - \vec{v}_q(k) \right)^2}$$

$$\Downarrow$$

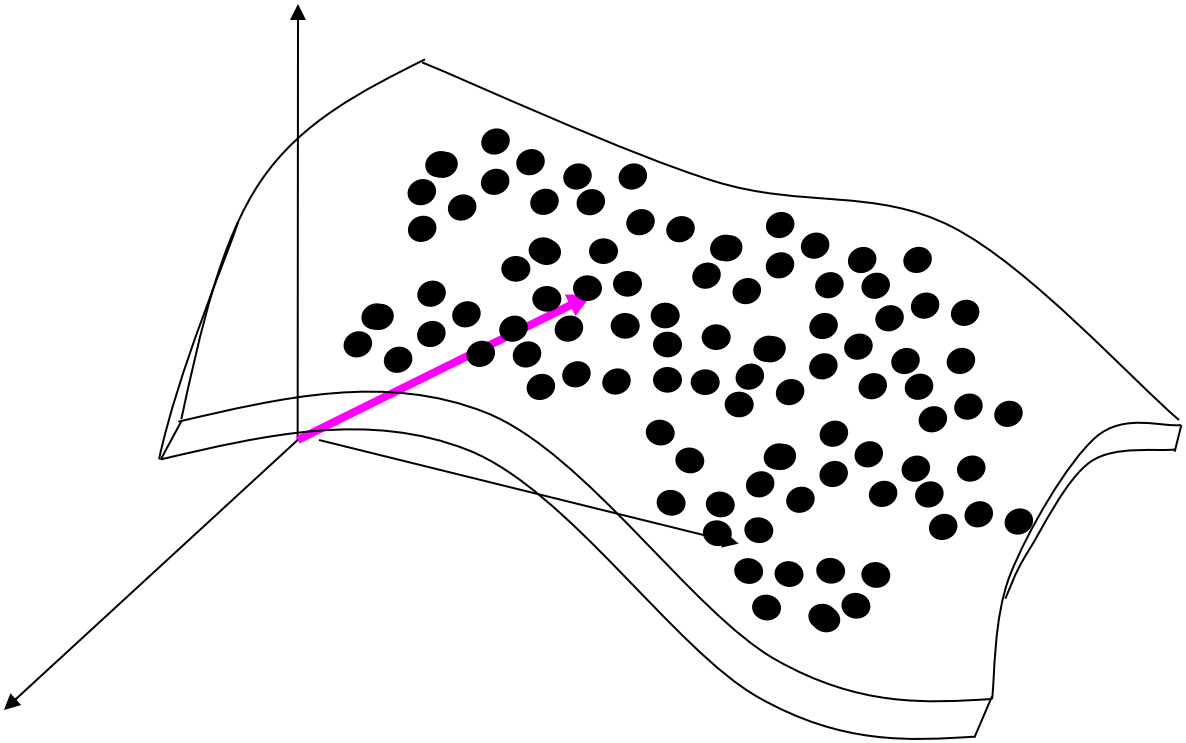
$$O(N^2) \quad operations / image$$



Cost of matching

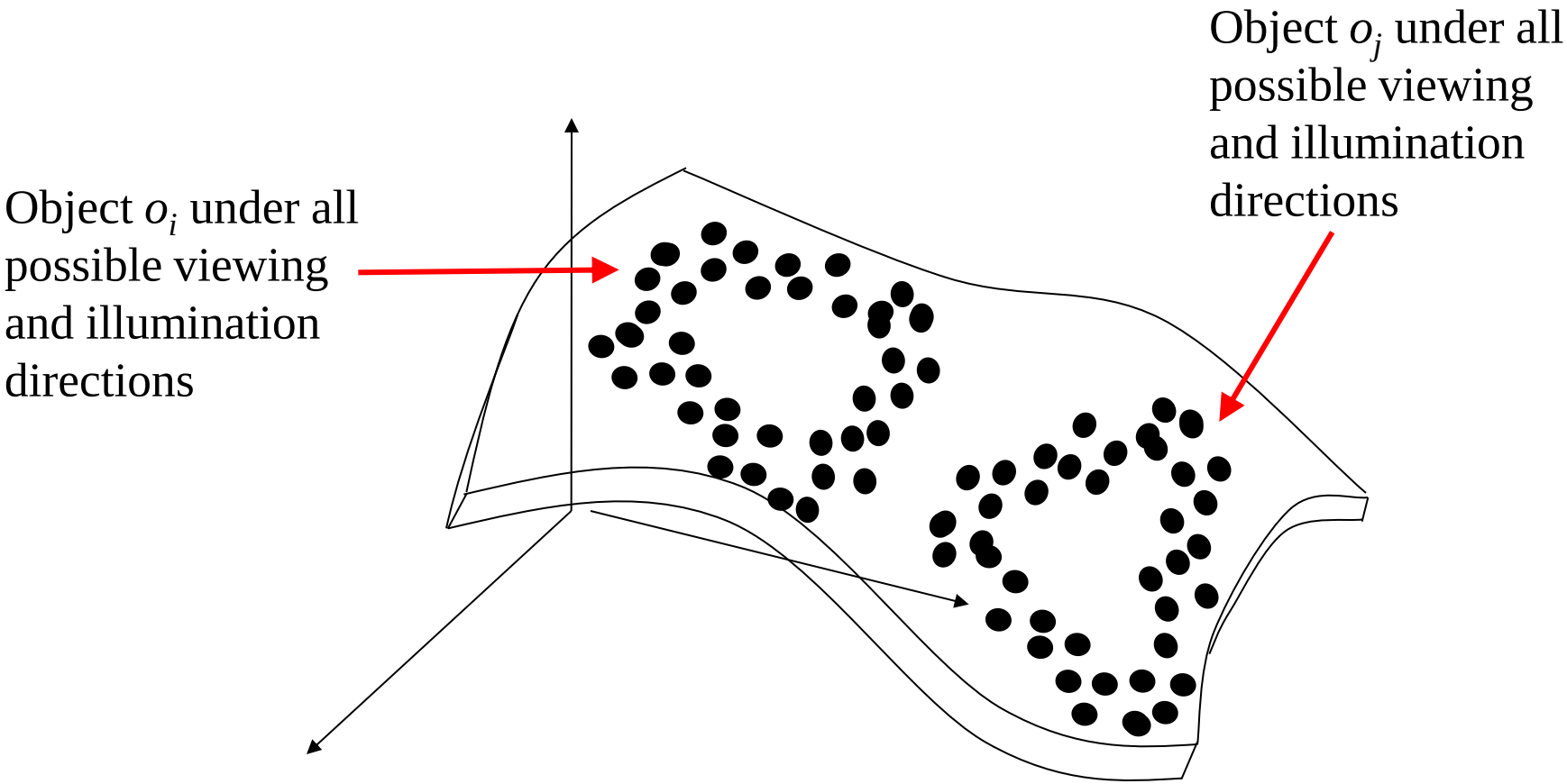
Appearance-Based Recognition

Image manifolds



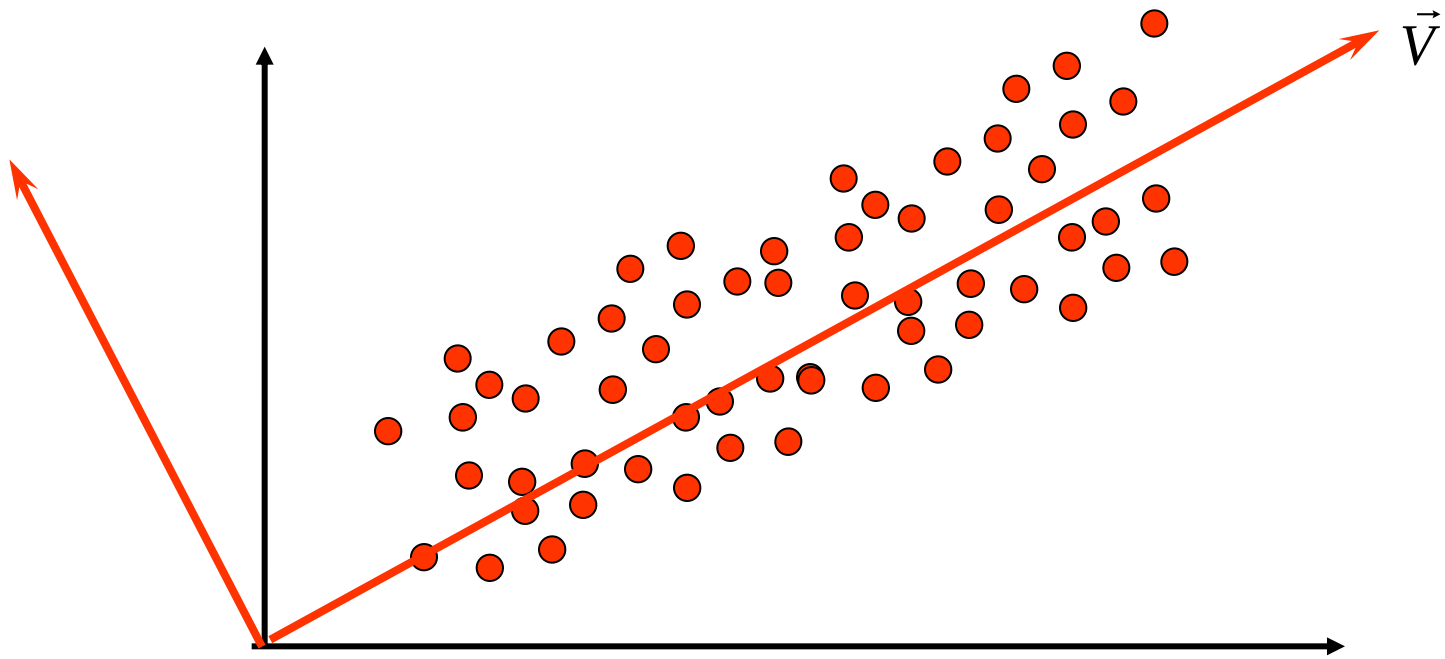
Appearance-Based Recognition

Object manifolds



Appearance-Based Recognition

Image manifolds and PCA



PCA: Find basis which maximizes directional variance

Appearance-Based Recognition

Eigenspace representation

Let $\vec{v}_1, \vec{v}_2, \ldots, \vec{v}_n$ be N^2-dimensional (image) vectors and $\bar{v} = \frac{1}{n}\sum_{j=1}^{n} \vec{v}_j$ their average vector. Let $\boldsymbol{X}$ be the $N^2 \times n$ matrix

$$X = \begin{bmatrix} | & & | \\ (\vec{v}_1 - \bar{v}) & \cdots & (\vec{v}_n - \bar{v}) \\ | & & | \end{bmatrix}$$

then each $\vec{v}_j$ can be written as

$$\vec{v}_j = \bar{v} + \sum_{i=1}^{n} g_{ji} \cdot \vec{\mathrm{e}}_i$$

where $\vec{\mathrm{e}}_1, \vec{\mathrm{e}}_2, \ldots, \vec{\mathrm{e}}_n$, the *principle components*, are the eigenvectors of the covariance matrix $Q = XX^T$ corresponding to the n (nonzero) eigenvalues of Q.

Appearance-Based Recognition

Eigenspace representation

$$\vec{v}_j \quad \Rightarrow \quad \vec{g}_j = (g_{j1}, \ldots, g_{jn})$$

Image representation in natural basis

Image representation in eigenspace

$$\lambda_1 \geq \lambda_2 \geq \ldots \geq \lambda_n \quad \Rightarrow \quad \vec{\mathrm{e}}_1, \vec{\mathrm{e}}_2, \ldots, \vec{\mathrm{e}}_n$$

Decreasing eigenvalues

Variance maximizing eigenvectors

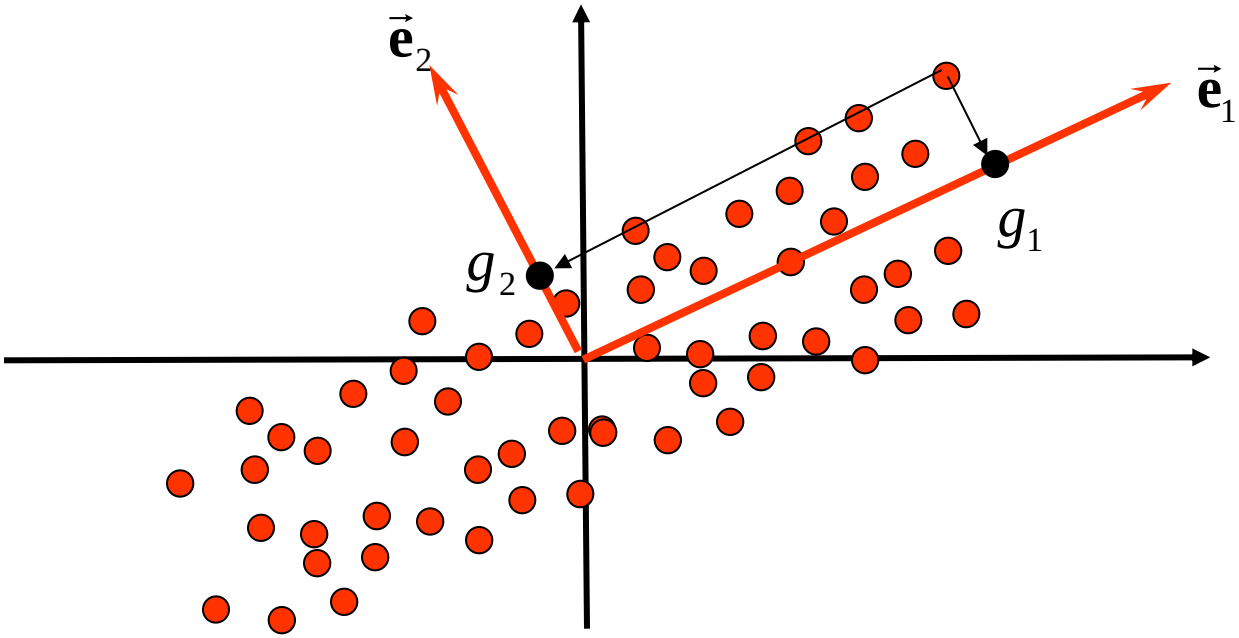
$$\lambda_{i>k} \approx 0 \quad \Rightarrow \quad \vec{v}_j \approx \bar{v} + \sum_{i=1}^{k} g_{ji} \cdot \vec{\mathrm{e}}_i$$

Vanishing eigenvalues

Compact (k-dim) image representation

Appearance-Based Recognition

Eigenspace and PCA



Appearance-Based Recognition

$O(N^2)$

Similarity in eigenspace

$$\left\|\vec{v}_j - \vec{v}_q\right\| = \left\|\left(\bar{v} + \sum_{i=1}^{n} g_{ji} \cdot \vec{\mathrm{e}}_i\right) - \left(\bar{v} + \sum_{i=1}^{n} g_{qi} \cdot \vec{\mathrm{e}}_i\right)\right\|$$

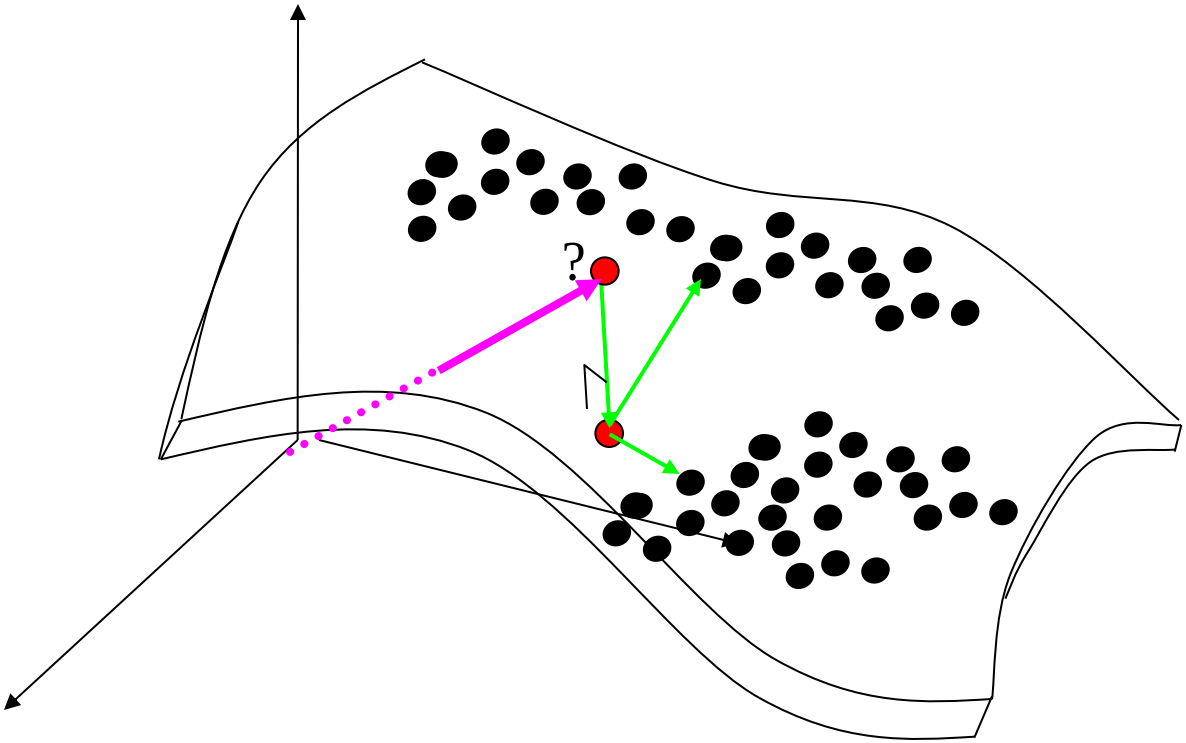
$$\approx \left\|\left(\bar{v} + \sum_{i=1}^{k} g_{ji} \cdot \vec{\mathrm{e}}_i\right) - \left(\bar{v} + \sum_{i=1}^{k} g_{qi} \cdot \vec{\mathrm{e}}_i\right)\right\|$$

$$= \left\|\sum_{i=1}^{k} (g_{ji} - g_{qi}) \cdot \vec{\mathrm{e}}_i\right\| = \sqrt{\sum_{i=1}^{k} (g_{ji} - g_{qi})^2} = \left\|\vec{g}_j - \vec{g}_q\right\|$$

$O(k)$

Appearance-Based Recognition

Recognition in the image eigenspace



Appearance-Based Recognition

Appearance-based recognition in the image eigenspace

1. Obtain a training set of images of all objects of interest under variable viewing and illumination conditions
2. Construct the image eigenspace from the training set using PCA
3. Given a query image
   - Project the image to the PCA subspace (eigenspace)
   - Compute distances to training set images in the eigenspace and find object maniford o_j with minimal distance d_{min}.
   - Unless d_{min} too large, identify query image with object o_j.

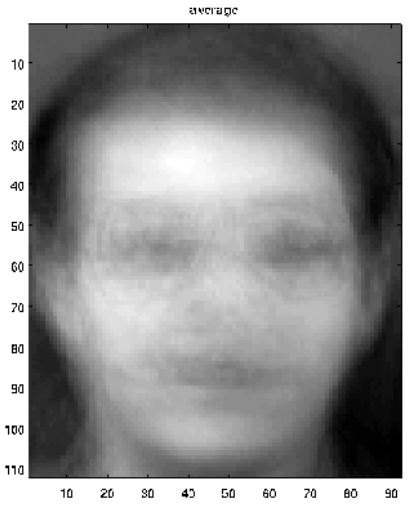
Appearance-Based Recognition

Example

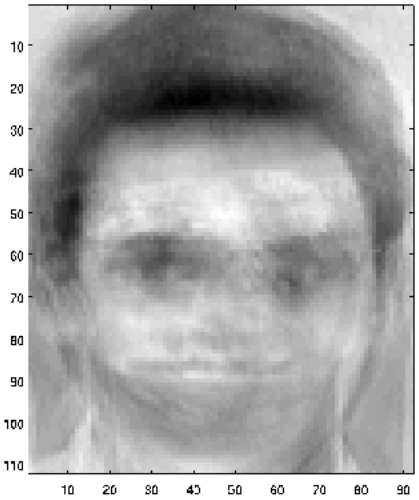
Database = 40 pictures, 10 individuals, 4 pictures per individual.

Appearance-Based Recognition

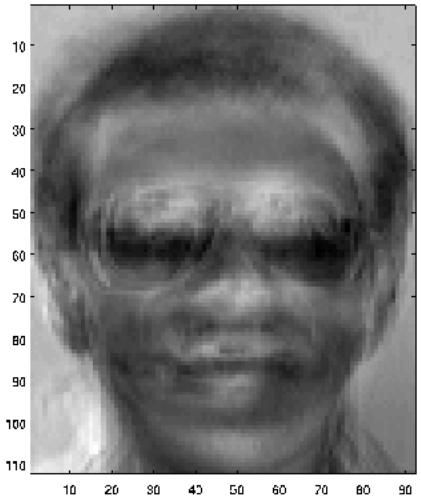
Example



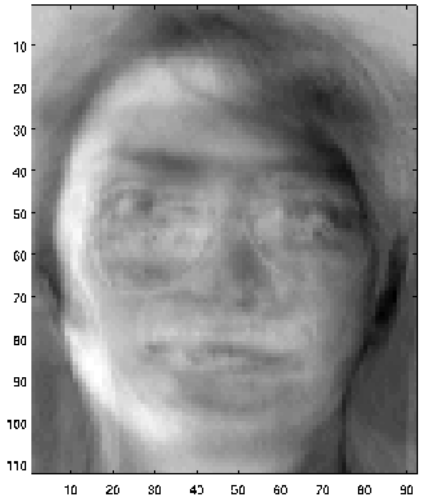
Average
"object"



$\vec{\mathbf{e}}_1$



$\vec{\mathbf{e}}_2$



$\vec{\mathbf{e}}_3$

First three principle components

Appearance-Based Recognition

Example

Query image (new)

Eigenspace reconstruction

(10 principle components)

Appearance-Based Recognition

Example